

Comparing the performance of SSD and HDD disks in a virtualized environment

Abstract — In modern IT systems, efficient management of storage resources is becoming crucial, especially in the context of virtualization. Traditional HDD drives, although reliable and more affordable, are increasingly falling behind SSD technology, which offers significantly higher data access speeds. However, the available literature rarely provides a clear picture of the actual performance differences between these two types of drives when used within virtual machines. This research aims to fill that gap by providing concrete data derived from systematic comparisons in a controlled virtualized environment.

The study was conducted using a Hyper-V virtual machine with identical hardware specifications for all tests. Two disk configurations were tested — SSD and HDD — using identical file systems (ext4, xfs, btrfs), and benchmarking tools Filebench and AS SSD Benchmark were employed. These tools allow for precise performance measurements under different workload types, including webserver, fileserver, varmail, and randomfileaccess scenarios, as well as sequential and 4K read/write operations.

The results clearly demonstrate a significant advantage of SSDs in nearly all categories. SSDs outperform HDDs in 4K and 4K-64Thrd operations and achieve up to ten times better performance in random data access. The effects of the file system are more pronounced with HDDs than with SSDs.

The conclusion of this paper is that the obtained results are directly applicable to the selection of storage resources for virtualized environments. The most important contribution is the empirical confirmation that SSDs significantly enhance system performance, especially under high I/O demand conditions, thus addressing the gaps in previous research that primarily focused on hypervisors.

Keywords: *Hyper-V, Fajl sistemi, Filebench, AS SSD Benchmark, Server scenariji, Performanse pristupa*

I. INTRODUCTION

The rapid development of information technologies and the growing reliance on virtual infrastructure services have made storage device performance a key factor in maintaining system stability and efficiency. Virtualization enables the simultaneous operation of multiple virtual machines on a single physical platform, leading to better resource utilization, cost reduction, and simplified IT environment management. However, this technology also introduces additional challenges, primarily regarding the performance of input/output (I/O) operations, which depend heavily on the characteristics of the underlying storage medium.

Traditional hard disk drives (HDDs), based on mechanical components, have served as the primary means of data storage for decades. However, their performance significantly degrades in scenarios involving frequent random access and multiple concurrent requests—conditions commonly encountered in virtualized environments. In contrast, modern solid-state drives (SSDs), based on NAND flash technology, offer incomparably higher access speeds, lower latency, and resistance to fragmentation. Although numerous commercial analyses and industry benchmarks have emphasized the superiority of SSDs, academic literature still lacks sufficiently systematic and empirically supported studies that uniformly compare the performance of SSD and HDD drives under virtualization conditions, across various workload types and file systems.

The contribution of this research lies in its systematic examination and comparison of SSD and HDD performance within a controlled virtualized environment using identical hardware and software parameters. A particular focus is placed on analyzing the impact of different workload types (sequential and random read/write operations) and file systems (ext4, xfs, btrfs) on the performance of both types of drives. The study employs standardized and widely accepted benchmarking tools: Filebench and AS SSD Benchmark, which enables replicability and objective comparability of results.

Within the Filebench testing, four real-world scenarios were analyzed: webserver (dominated by random reads), fileserver (mixed I/O patterns), varmail (synchronous writing of small files), and randomfileaccess (scattered read/write of various files).

Meanwhile, AS SSD Benchmark was used to evaluate the following categories: sequential read and write (SEQ), 4K read/write, as well as 4K access with 64 parallel threads (4K-64Thrd), for both operations — read and write.

Existing studies focusing on virtualization performance most often concentrate on comparing different hypervisors — such as ESXi, Xen, and KVM — but frequently overlook the detailed impact of physical storage devices on overall performance. Additionally, studies like those conducted by Kamar & Shiwani (2014) or Polenov & Guzik (2018) tend to rely on generic tools such as Bonnie++ or Iozone, without analyzing the context of specific workloads and file systems. For this reason, there is a clear need for more targeted and methodologically rigorous research, which this study aims to fulfill.

Through this work, the study not only addresses the practical challenge of optimizing storage systems in virtual environments but also fills an important gap in the academic literature regarding the concrete impact of disk type selection on system performance, depending on usage scenarios and the software layer of the file system.

II. RELATED WORK, OBJECTIVE AND MOTIVATION

In previous studies on virtualization, attention has primarily been focused on comparing the performance of hypervisors such as VMware ESXi, KVM, and Xen, while storage devices—particularly HDD and SSD drives—have often been overlooked [1]. For example, White and Pilbeam analyzed the overall impact of virtualization on system performance but without a detailed breakdown of I/O layer performance [1]. Similarly, Mandebi Mbongue and collaborators compared virtualization with containerization, focusing mainly on CPU and network traffic, rather than on the storage medium [2].

More advanced architectural innovations, such as the File System De-Virtualizer (FSDV) introduced by Zhang and colleagues, highlight the benefits of removing virtualization layers when working with SSDs; however, their analysis was largely theoretical and did not include a direct comparison with HDDs in a virtual machine environment [3]. On the other hand, Nguyen provided valuable insight into the predictability of SSD I/O performance in virtualization, confirming that

under performance-intensive workloads, SSDs may exhibit less predictable performance behavior compared to HDDs [4].

Existing research also addresses hybrid storage solutions and caching mechanisms, but again without directly comparing SSD and HDD drives within the same virtualized machine [5]. While there is a clear industry trend toward adopting SSDs for speed improvements, factors such as cost considerations, file system impact, and workload type remain insufficiently explored.

The motivation for this study stems from the need to explicitly investigate how SSD versus HDD performance manifests in real I/O scenarios within virtual machines, using standardized tests (webserver, fileserver, varmail, randomfileaccess in Filebench, and SEQ, 4K, 4K-64Thrd in AS SSD Benchmark). The goal is to overcome the limitations of existing works, which either treat storage as a secondary factor or fail to analyze the influence of file systems and various workload types in depth..

III. Architecture of HDD and SSD drives

Storage devices represent a critical component in any computer system, as they directly affect the performance of input/output (I/O) operations. Traditionally, the Hard Disk Drive (HDD) has been the dominant data storage technology, but in the past decade, Solid State Drive (SSD) technology has gained increasing interest and adoption. Understanding their architectures is essential for the correct interpretation of performance differences, especially in virtualized environments [6].

An HDD is an electromechanical device that uses magnetic platters coated with ferromagnetic material to store binary data. These platters rotate at high speeds—typically 5400, 7200, or 10,000 revolutions per minute (RPM)—while a read/write head, mounted on an actuator arm, hovers above them [7]. The actuator arm allows precise positioning of the head over the desired sector of the disk.

Data is stored in circular tracks and sectors, organized in a cylindrical structure. The read/write head must physically move to the point where the data resides, introducing latency, particularly in scenarios involving random access. Additionally, seek time and rotational latency are key limiting factors in HDD I/O performance [8], [9].

HDDs use interfaces such as SATA or SAS, and they typically include an internal cache ranging from 16 MB to 256 MB, which is used for temporarily storing frequently accessed data. Although HDDs are well-suited for storing large volumes of data at a low cost, their mechanical nature makes them more prone to physical failure and significantly slower in demanding I/O operations.

SSDs, by contrast, operate on a completely different principle—rather than mechanical parts, they utilize electronic components, primarily NAND flash memory chips, for permanent data storage. Data is read and written through a controller, which manages access to memory blocks and performs tasks such as caching, wear leveling, compression, and error correction (ECC) [10].

Unlike HDDs, SSDs allow instant access to any memory block without the need for physical movement. This feature greatly reduces latency and enables much higher read/write speeds, especially in scenarios involving random access to small files (e.g., 4K I/O blocks) [11].

SSDs commonly use interfaces such as SATA, NVMe (via PCIe bus), or M.2. Modern SSDs support multi-channel architecture, allowing the controller to communicate with multiple NAND chips simultaneously [12]. This parallelism enhances bandwidth utilization and results in significantly improved overall performance.

Table I presents a comparison of HDD and SSD architectures.

TABLE I. HDD AND SSD COMPARASION

Characteristic	HDD	SSD
Technology	Mechanical, magnetic plate	Electronic, flash memory
Movable discs	Yes	No
Latency	High (seek time + rotational)	Low (electronic access)
Sequential speed	Up to 200 MB/s	500 MB/s (SATA), 3+ GB/s (NVMe)
Random I/O operation	Limited (high seek time)	Excellent (quick access to small blocks)
Durability (writing)	Very high	Limited number of write cycles
Impact resistance	Low	High
Price per GB	Low	High

This architectural difference makes SSDs more suitable for applications with high I/O demands, such as databases, virtual machines, and hosting servers, while HDDs remain a viable solution for archiving and backup purposes.

IV. File system

The file system represents a crucial component of every operating system, as it determines how data is organized, indexed, and accessed on a physical or virtual disk. The performance of storage devices such as SSDs and HDDs significantly depends on the efficiency and functionality of the selected file system. This study examines three modern Linux file systems — ext4, XFS, and Btrfs — in order to investigate their interdependence with device type and workload [13].

ext4 is an improved version of ext3 and is considered one of the most stable and widely used file systems in Linux environments. Its reliability, low overhead, and broad compatibility make it the default choice in many Linux distributions.

Key technical features of ext4 include:

- Delayed Allocation – a technique that postpones physical data mapping to disk in order to optimize block allocation and reduce fragmentation.
- Extent-based Allocation – unlike traditional block-based allocation, it uses “extents” (contiguous block regions), enabling more efficient access to large files.
- Journal Checkpointing – employs ordered journaling, where data is written first, followed by metadata, reducing the risk of data corruption..

ext4 delivers stable and predictable performance, particularly on HDDs where timing characteristics such as seek time and latency are mitigated by linear data layout. On SSDs, ext4 uses less sophisticated optimization mechanisms; although it still performs well, it does not fully utilize the parallelism and unique characteristics of flash memory.

XFS is a file system designed with scalability and high throughput in mind, originally developed by Silicon Graphics and now widely supported in the Linux kernel. It is most commonly used in environments that require handling large files and parallel I/O operations.

Its distinctive technical components include:

- B+-tree indexing of directories and block allocation, allowing for fast search and modification.
- Pre-allocation and Asynchronous Journaling – transactions are logged independently from the main I/O stream, improving access speed.
- Striping support – enables optimization for RAID configurations.

On HDDs, XFS performs stably during sequential operations on large files but may introduce higher overhead with many small files. On SSDs, its ability to handle multiple I/O requests simultaneously becomes prominent, especially in multi-threaded tests such as 4K-64Thrd, where it demonstrates consistent and scalable performance.

Btrfs is the most advanced of the file systems analyzed and was designed as a long-term replacement for ext4. Its architecture is based on B-trees, allowing for complex functionalities such as snapshots, incremental backups, integrated RAID support, and data integrity verification.

Technical characteristics include:

- Copy-on-Write (CoW) – every data modification creates a new instance, enabling snapshot capabilities and preventing overwriting of active data.
- Checksumming – integrated data integrity protection through CRC32C hash functions.
- Transparent compression – supports zlib, lzo, and zstd algorithms..

On SSDs, the CoW mechanism enables efficient data write distribution and reduces excessive writing, which is beneficial due to the limited number of program/erase (P/E) cycles in flash memory. However, high CPU consumption and complex metadata handling may negatively impact performance with small files and high I/O frequency. On HDDs, the structural complexity and fragmentation result in weaker responsiveness compared to ext4 and XFS.

V. HYPOTHESES OF EXPECTED BEHAVIOR

In order to systematically compare the performance of SSD and HDD drives in a virtualized environment, several hypotheses have been formulated to represent the expected system behavior based on theoretical foundations, previous research, and the physical characteristics of the analyzed devices. Each hypothesis is based on measurable parameters that will be tested using standardized tools and realistic workload scenarios.

- 1) Hypothesis H1: SSD will outperform HDD multiple times in random access tests (4K and 4K-64Thrd)

Random I/O operations involving small data blocks (typically 4 KB in size) represent the type of workload that

most clearly exposes the architectural limitations of traditional HDD devices. In HDDs, the total access time (latency) depends on several physical factors:

$$T_{sum} = T_{seek} + T_{rotation} + T_{transfer}$$

- T_{seek} : head positioning time over the corresponding track (typically 3-10 ms)
- $T_{rotation}$: waiting time for the sector to rotate under the head (eg 4.17 ms for 7200 rpm)
- $T_{transfer}$: only data transfer time, in μs

For SSDs, which do not rely on mechanical components, the total latency is on the order of a few microseconds ($<100 \mu s$), providing a significant advantage in 4K and 4K-64Thrd tests, where the number of read/write operations involving small data blocks is measured under high levels of concurrent access.

$$IOPS_{SSD}^{4K} \gg IOPS_{HDD}^{4K} \quad i \quad IOPS_{SSD}^{64Thrd} \gg IOPS_{HDD}^{64Thrd}$$

- 2) Hypothesis H2: SSD will have significantly higher sequential read and write speeds (SEQ) compared to HDD

In sequential data access, blocks are read in order from contiguous sectors. HDDs achieve their best performance under these conditions because seek time and rotational latency are minimized. However, SSD devices, utilizing parallel access (multiple NAND channels plus DRAM cache), achieve significantly higher transfer speeds, which are limited primarily by the interface (SATA III ~550 MB/s; NVMe >3 GB/s).

The measured parameters will be:

- Sequential reading speed B_{read}^{seq}
- Sequential write speed B_{write}^{seq}

Expected:

$$B_{SSD}^{seq} \gg B_{HDD}^{seq}$$

Where B is the bandwidth expressed in MB/s.

- 3) Hypothesis H3: The XFS file system will provide the best results on an SSD in tests using random I/O

The XFS file system employs extent-based allocation algorithms and B+-tree indexing, making it well-suited for parallel operations. Its architecture supports multiple concurrent I/O request streams, which aligns well with the internal parallelism of SSD devices. In tests such as randomfileaccess or 4K-64Thrd, XFS is expected to leverage SSD parallelism more efficiently than ext4 or Btrfs.

Parameter: Average throughput of I/O operations in multithreaded workloads $P_{FS,SSD}^{multi}$

Expected:

$$P_{xfs}^{SSD} > P_{ext4}^{SSD} > P_{btrfs}^{SSD}$$

- 4) Hypothesis H4: ext4 file system will show the best balance between stability and performance on HDD

ext4 was developed as a stable and efficient file system with low overhead and performs exceptionally well in systems with limited I/O capacity. Its journaling mechanism and delayed allocation strategy provide advantages in linear and mixed workloads. On HDD devices, where every additional layer of complexity increases overhead, ext4 is expected to deliver better and more predictable results.

Expected:

$$P_{ext4}^{HDD} > P_{xfs}^{HDD} \approx P_{btrfs}^{HDD}$$

during tests with a large number of smaller files (fileservers, varmail).

- 5) Hypothesis H5: Btrfs will have the highest overhead and lowest performance on HDD in all categories

Due to the complex Copy-on-Write logic, internal snapshot mechanisms, and checksum generation, Btrfs incurs significant CPU and I/O overhead. While this overhead can be somewhat mitigated on SSD drives, on HDDs—which have higher latency and lower throughput—such software demands further degrade performance.

$$P_{btrfs}^{HDD} \ll P_{ext4/xfs}^{HDD}$$

Especially with random write scenarios and workloads like varmail and randomfileaccess.

- 6) Hypothesis H6: Differences between file systems will be more pronounced on HDD than on SSD

For SSD drives, hardware advantages such as lower latency, higher IOPS, and DRAM caching enable effective abstraction from the file system. In contrast, for HDDs, differences in file system allocation algorithms and data access methods directly impact performance.

Expected:

$$\Delta P_{FS}^{HDD} > \Delta P_{FS}^{SSD}$$

where is: $\Delta P_{FS}^x = \max(P_{FS1}^x - P_{FS2}^x)$, performance difference between file systems FS1 and FS2 on disk type $x \in \{HDD, SSD\}$

VI. TEST CONFIGURATION AND TOOLS USED

The goal of this experimental section was to systematically and reproducibly test the impact of storage device type (SSD versus HDD) on performance in a virtualized environment, under various workload conditions and using multiple file systems. Given the complexity of I/O operations in modern systems, the testing was conducted under strictly controlled environmental parameters to ensure the validity and consistency of the results.

All tests were performed on a virtual machine created on a Hyper-V hypervisor, installed on a host system with appropriate hardware resources (see Table II). The virtual machine was configured as shown in Table III.

TABLE II. HOST MACHINE

Components	Characteristic
CPU	AMD Ryzen 5 5600H Zen 3, 2021
CPU clock	3.3 Ghz
CPU cores	6
CPU cache	L3 16 MB
RAM	2x8 GB DDR4, 3200 MHz
Storage	SSD 512 GB NVMe

TABLE III. VIRTUAL MACHINE

Components	Characteristics
vCPU	3
Memory	4 GB
SSD	1 partition 60 GB /dev/sda root
Sdb1	Test partition for SSD – 30 GB
Sdc1	Test partition for HDD .- 30 GB

Although direct access to the physical disk via Hyper-V passthrough functionality is not used, this configuration still enables reliable testing of performance differences, as the .vhdx files are physically separated and mapped to distinct storage devices. This eliminates the effect of shared storage and ensures comparability of results. The specifications of the SSD can be found in Table IV, while those for the HDD are listed in Table V.

TABLE IV. SSD TEST ENVIRONMENT

WDC PC SN530 SDBPMPZ-512G-1101	
Components	Characteristics
Cache memory	None (uses HMB - Host Memory Buffer)
Sequential data read speed	2500 MB/s
Sequential data write speed	1770 MB/s
IOPS	315K/230K iops
Interface	PCIe Gen 3 x4, NVMe

TABLE V. HDD TEST ENVIRONMENT

WDC WD5000LPVX-22V0TT0	
Component	Characteristics
Cache memory	8 MB
Number of revolutions RPM	5400 RPM
Seek times	12ms
STR (internal media rate)	150 MB/s
Interface	SATA III

Before each test set, every disk was formatted with the selected file system: ext4, XFS, or Btrfs. ext4 was chosen as the reference file system due to its stability and low overhead, XFS for its strong support for parallel access and large file handling, and Btrfs was included in the tests because of its advanced features such as snapshots and the Copy-on-Write (CoW) mechanism, which affect SSD and HDD devices differently.

Two tools were used to measure performance: Filebench and AS SSD Benchmark. Filebench was utilized to simulate realistic workloads using predefined profiles: webserver (read-intensive access to small files), fileserver (mixed read/write patterns), varmail (frequent writing of small files), and randomfileaccess (random access to files of varying sizes).

Each test was executed three times, and between each run the system's memory cache was manually cleared (`sync && echo 3 > /proc/sys/vm/drop_caches`) to eliminate the influence of previous I/O operations.

Additional measurements were conducted using AS SSD Benchmark, which evaluates throughput and IOPS under different access patterns. Tests included: sequential read and write (SEQ), random 4K operations, and 4K access with 64 parallel threads (4K-64Thrd), both for reading and writing.

These tests were executed with caching disabled and all background services turned off, to ensure maximum accuracy and consistency of the results.

All tests were performed in minimal system mode, without a graphical interface and with no additional active services, in order to maximize control over system load. For all measurements, the average value of the three consecutive test runs was used, and it was verified that the variance did not exceed the allowed threshold ($\pm 5\%$). This methodological approach provided analytical precision and a reliable foundation for further performance comparison between SSD and HDD devices under varying conditions.

VII. TEST RESULTS AND ANALYSIS

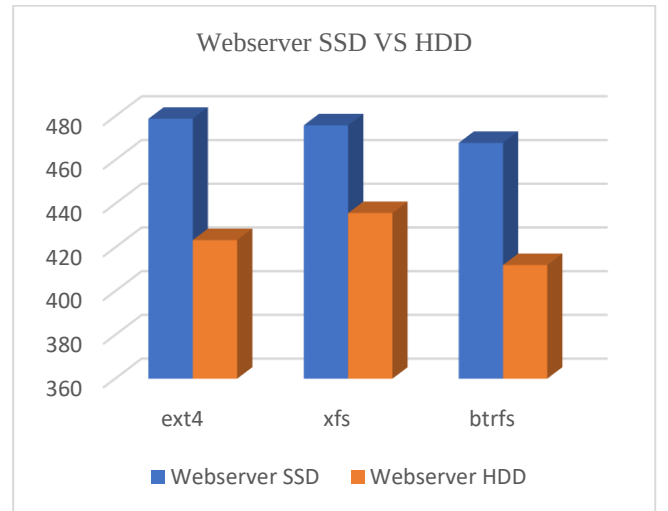
The results of the experimental testing represent the central part of this research and serve to verify the stated hypotheses. Performance was measured using two complementary tools — Filebench, which simulates realistic workloads, and AS SSD Benchmark, which provides low-level analysis of I/O performance under different access patterns [14]. Each test was conducted on both SSD and HDD drives, using three different file systems: ext4, XFS, and Btrfs. The collected data is presented in tabular and graphical formats, followed by analysis based on the type of workload.

Filebench testing included four different workload profiles. The webserver workload simulates a read-intensive access pattern to a large number of static files, typical for server applications such as Apache or Nginx. The results are shown in table VI and illustrated in figure I. In this test, the SSD performed slightly better than the HDD, although the differences were not substantial — which is expected, as the workload involves mostly sequential reads with a low number of write operations. Among the file systems tested on the SSD, ext4 achieved the highest throughput (478.90 MB/s), followed by XFS (475.90 MB/s), while Btrfs showed the lowest value (467.87 MB/s). On the HDD, a similar pattern was observed, with XFS yielding the best result (435.77 MB/s), suggesting its efficiency in handling linear operations.

TABLE VI. FILEBENCH WEBSERVER TEST RESULTS

Webserver		
Filesystem	SSD	HDD
ext4	478.9	423.4
xfs	475.9	435.77
btrfs	467.87	411.97

Figure I. FILEBENCH WEBSERVER TEST RESULTS

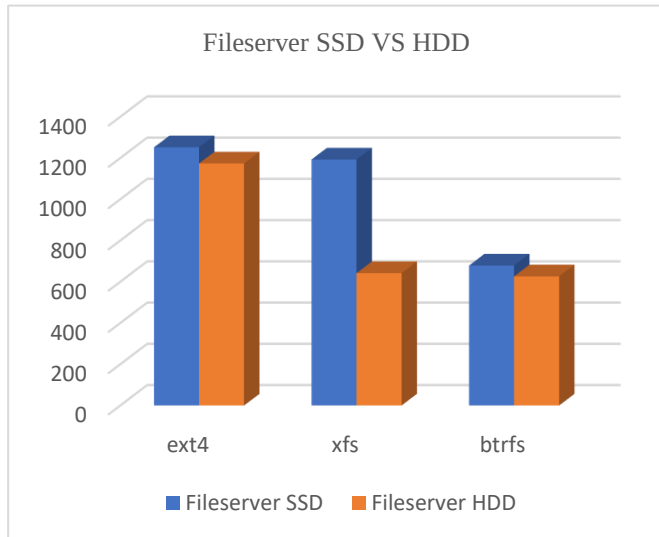


In the fileserver workload, which combines reading and writing files of varying sizes, a more noticeable drop in performance was observed on the HDD when using the XFS and Btrfs file systems. The results are presented in table VII and illustrated in figure II. On the SSD, ext4 achieved the highest throughput (1252.17 MB/s), followed by XFS with slightly lower performance (1192.73 MB/s), while Btrfs showed significantly reduced performance (678.77 MB/s). On the HDD, ext4 again led the results (1173.47 MB/s), whereas XFS dropped to 641.77 MB/s, and Btrfs to 626.17 MB/s. This confirms the hypothesis that ext4 offers the best balance of performance on HDDs, while Btrfs incurs high overhead under mixed I/O conditions.

TABLE VII. FILEBENCH FILESERVER TEST RESULTS

Fileserver		
Filesystem	SSD	HDD
ext4	1252.17	1173.47
xfs	1192.73	641.77
btrfs	678.77	626.17

Figure II. FILEBENCH FILESERVER TEST RESULTS

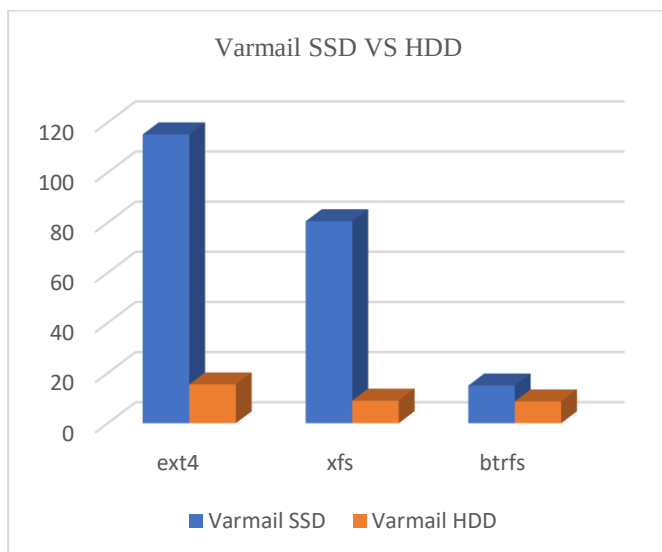


The Varmail workload tests system behavior under conditions that involve frequent writing of small files—typical for server applications that handle a large number of user messages. The results are presented in table VIII and illustrated in figure III. This test clearly demonstrates the superiority of SSD devices. On the SSD, ext4 reaches 115.20 MB/s, while XFS and Btrfs perform significantly worse (80.57 MB/s and 15.03 MB/s, respectively). The HDD exhibits extremely low values, particularly for Btrfs (8.70 MB/s), which confirms its vulnerability in write-intensive scenarios with a high number of small transactions.

TABLE VIII. FILEBENCH VARMAIL TEST RESULTS

Varmail		
Filesystem	SSD	HDD
ext4	115.2	15.47
xfs	80.57	9
btrfs	15.03	8.7

Figure III. FILEBENCH VARMAIL TEST RESULTS



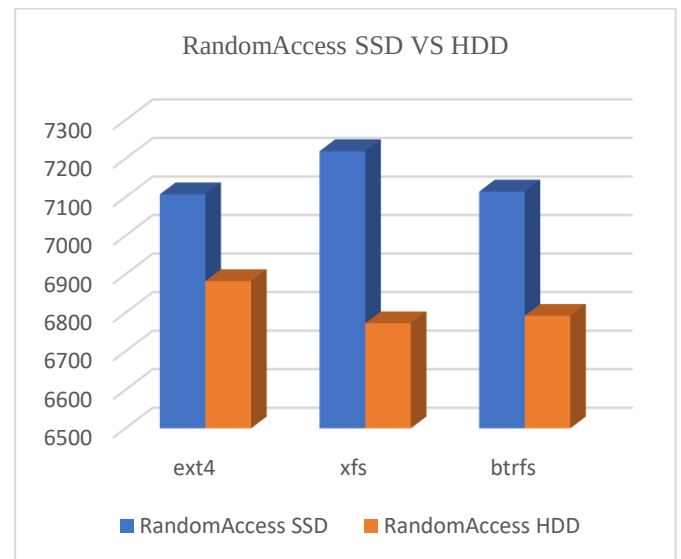
The randomfileaccess workload measures system behavior during reading and writing of files with random sizes and locations. This test is most relevant for evaluating latency and IOPS performance. The results are presented in table IX and illustrated in figure IV. On the SSD, XFS achieved the best result (7219.23 MB/s), while ext4 (7107.27 MB/s) and Btrfs (7114.43 MB/s) were very close behind. Interestingly,

the HDD also showed high values across all file systems (above 6700 MB/s), which may indicate the effect of aggressive caching in virtual disk drivers or deferred writes. Nevertheless, the SSD demonstrates a clear advantage in performance consistency in this test.

TABLE IX. FILEBENCH RANDOMACCESS TEST REUSLT

RandomAccess		
Filesystem	SSD	HDD
ext4	7107.27	6882.03
xfs	7219.23	6772.83
btrfs	7114.43	6792.07

Figure IV. FILEBENCH RANDOMACCESS TEST RESULTS

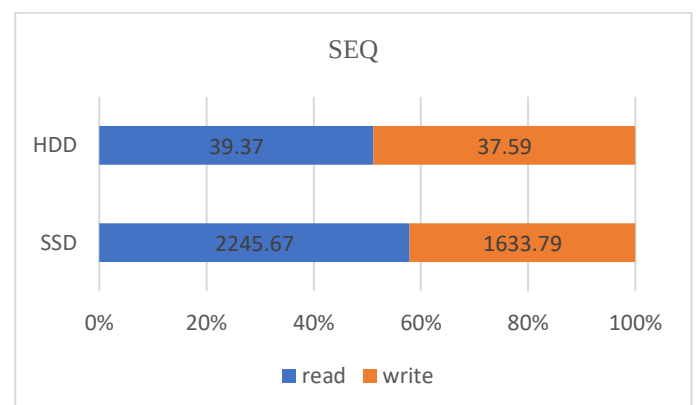


The results from the AS SSD Benchmark test further confirm the performance differences between SSD and HDD devices. In sequential access (SEQ), the SSD achieves 2245.67 MB/s for read and 1633.79 MB/s for write, while the HDD reaches only 39.37 MB/s for read and 37.59 MB/s for write, as shown in table X and figure V. This demonstrates the substantial advantage of SSDs and supports Hypothesis H2 regarding sequential operations.

TABLE X. AS SSD SEQ TEST RESULTS

SEQ	SSD	HDD
read	2245.67	39.37
write	1633.79	37.59

Figure V. AS SSD SEQ RESULTS



Particularly notable are the differences in 4K random access—a test that most accurately reflects behavior under small, random I/O requests. The SSD reaches 40.94 MB/s for 4K read and 128.42 MB/s for 4K write, while the HDD records only 0.46 MB/s and 1.45 MB/s, respectively, as presented in table XI and figure VI. An even more pronounced contrast is observed in the 4K-64Thrd test, where the SSD achieves 1070.72 MB/s for reading and 1243.59 MB/s for writing, while the HDD remains at nearly unusable levels of 0.58 MB/s and 1.4 MB/s, as shown in Table XII and Graph VII. These results confirm Hypothesis H1 and clearly illustrate that SSDs are incomparably more efficient under conditions involving multithreaded access and small block sizes.

TABLE XI AS SSD 4K TEST RESULTS

4K	SSD	HDD
read	40.94	0.46
write	128.42	1.45

Figure VI. AS SSD 4K TEST RESULTS

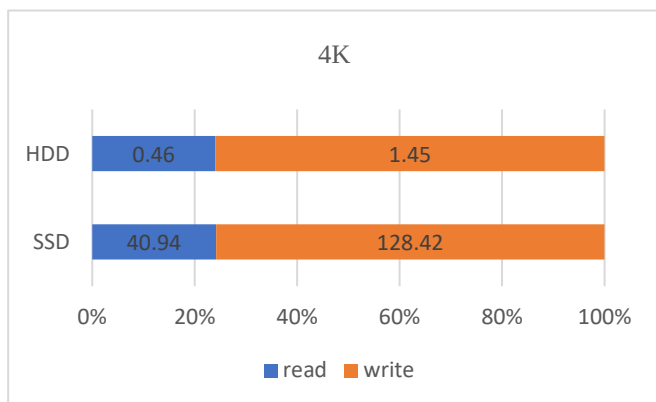
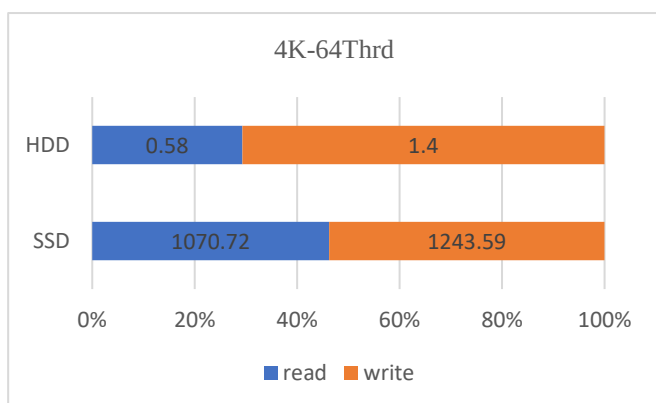


TABLE XII. AS SSD 4K-THRD TEST RESULTS

4K-64Thrd	SSD	HDD
read	1070.72	0.58
write	1243.59	1.4

Figure VII. AS SSD 4K-THRD TEST RESULTS



Based on the analysis of all obtained results, it can be concluded that SSD devices overwhelmingly outperform HDDs across all measured parameters. The ext4 file system proved to be the most stable and best-suited for mixed workloads, while XFS demonstrated the highest efficiency in random access scenarios on SSDs. Btrfs, although feature-rich, showed weaker performance in almost all scenarios, especially on HDDs, which supports Hypotheses H3, H4, H5, and H6. The performance differences between file systems are more pronounced on HDDs, confirming that the

underlying storage architecture and file system layer must be considered together when designing virtualized services.

VIII. CONCLUSION

As part of this research, a comprehensive analysis of SSD and HDD performance in a virtualized environment was conducted using various file systems and by simulating realistic workloads. Based on the executed tests, measured metrics, and compared throughput, it can be concluded that the hypotheses established prior to the experiment were largely confirmed, while certain nuances revealed additional factors that influence the behavior of storage systems.

The most clearly confirmed hypothesis relates to the drastic superiority of SSD devices in tests involving random access to small data blocks. Especially in multithreaded scenarios (4K-64Thrd), the SSD achieved performance levels over a thousand times higher than the HDD, thereby unequivocally confirming Hypothesis H1. This difference was particularly evident in server-like workloads such as varmail and randomfileaccess, where IOPS and latency are critical system factors.

Hypothesis H2, which pertains to sequential data access, was also fully confirmed. The SSD achieved sequential read and write speeds more than 50 times higher than the HDD, indicating that even under optimal conditions for the HDD, SSDs maintain both a hardware and logical advantage due to multi-channel architecture and the absence of mechanical delays.

Regarding file systems, Hypothesis H4 was confirmed — ext4 proved to be the most stable and optimal file system in most tests on HDD. XFS, consistent with Hypothesis H3, achieved slightly better results in random access tests on SSDs, while Btrfs emerged as the least efficient under high I/O demands, especially on HDDs, reinforcing Hypothesis H5.

Hypothesis H6, which assumes that differences between file systems would be more pronounced on HDDs, also proved to be accurate. On SSDs, the advanced architecture tends to mitigate software-level differences, whereas on HDDs, the device reacts more sensitively to data distribution and allocation strategies used by the file system. Changes in file systems on HDDs resulted in greater variations in throughput and latency than on SSDs, confirming the importance of their interdependence.

Considering all the analyzed data, it can be concluded that the choice of storage device has a decisive impact on performance in virtualized environments, but software factors such as file system type and workload pattern further modulate this effect. SSDs are unquestionably superior in nearly all tested scenarios, with performance differences that are quantitatively measurable and practically relevant for the implementation of server solutions, development of virtualized environments, and infrastructure planning in domains where I/O performance is critical.

For future research, it is recommended to include passthrough disk configurations to eliminate virtualization layers from the I/O path, as well as to incorporate NVMe SSD technology and conduct a deeper analysis of high-level file system features, such as journaling modes, write-back caching, and TRIM support. Additionally, analyzing durability and energy consumption under similar conditions could provide further insight into the overall performance of storage solutions in virtualized systems

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